Auditable Exact Lower Bounds for Trajectory QCQPs via a Rational Riccati Sweep

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Abstract

Trajectory optimization with keep-out state constraints gives nonconvex quadratically constrained quadratic programs (QCQPs). A feasible trajectory supplies an upper bound on the optimum; checking a global lower bound requires a separate certificate. We study a subclass with quadratic penalties between neighboring stage states and spherical keep-outs at those states.

For a supplied rational multiplier vector, the constant-multiplier S-procedure reduces certification to a bordered positive-semidefiniteness test. We prove that a scalar band-$LDL^\top$ sweep carries the affine border, handles singular pivots through an exact range test, and computes the Lagrangian lower bound in $O(Nd^3)$ rational operations for N stages of dimension d. Its positive-definite block form is the classical Riccati / block-Schur recursion. The bound is computed independently of the feasible trajectory; neither a rational optimum nor exact primal stationarity is needed. Indefinite curvature or a failed range test causes refusal.

Under aggregate-PSD exactness, rational contact and independent active gradients, the corresponding rational KKT pair closes the bracket. Constructed closing families serve as exact test oracles; three selected numerical-proposal examples exercise the general protocol, including irrational contact and a positive relaxation gap. A companion formalizes the acceptance predicate in Lean, but not the implementation. Current implementations use dense storage. The certificate concerns the rational problem at its sampled states; it does not establish safety between samples or certify the underlying continuous spacecraft dynamics.

1 Introduction

Guidance for spacecraft rendezvous, proximity operations, and docking (RPOD) plans a trajectory by solving optimal-control problems that, after discretization, are QCQPs: a quadratic cost (tracking, path smoothness) subject to quadratic keep-out constraints such as obstacles and abort-safety balls. Corridors and other convex tubes carry the opposite sign and are not instances of the class treated here. We treat throughout the *penalty-coupled* (soft-dynamics) subclass, in which the stage-to-stage dynamics enter the cost rather than as equality constraints and the decision variables are the stage states alone. Unconstrained controls with rational positive-definite quadratic cost and rational invertible square input maps can be eliminated into this form; general hard equality dynamics require a different formulation (§6).

These QCQPs are nonconvex, and standard practice — sequential convex programming, or numerical moment–SOS relaxations [23, 21, 43] — returns a solution whose global optimality is either not certified at all or certified in floating point, up to a tolerance. Lossless convexification [1]

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does certify exactly and analytically, via the maximum principle, but for nonconvex *control*-set constraints — not the state constraints at issue here.

We ask a different question: can the *certificate* of global optimality of a trajectory QCQP be made *exact*, checked in rational arithmetic with no tolerance, so that acceptance is a theorem about the deployed numbers rather than a numerical near-miss? For a single keep-out the answer is known — the S-lemma is exact and its dual is one PSD test [34, 42], with rational recovery [33, 20]. For a chain of coupled primitives, we study the precise verification question: how can the band structure be preserved while carrying the affine border and treating singular curvature exactly? A level-one relaxation gap may prevent closure, and general QCQPs can require exponential-bit rationals [6]; neither issue is resolved by a faster PSD test. The question is about checking, not searching. A proposer supplies the candidate — the numerical solvers of §4 or the inverse design of §5 — and nothing in the certificate route replaces the search that produced it.

The structural observation. The S-procedure certificate for the chain is one bordered PSD condition $M(\lambda, t) \succeq 0$ (§2). Its exact test is a rational $LDL^\top$ that eliminates within the problem's band and carries the affine column alongside; that test is precisely the discrete Riccati / block-Schur recursion of LQ optimal-control numerics [41, 35] where the block pivots stay positive, with the scalar band-$LDL^\top$ sweep carrying the same verdict across singular and indefinite pivots elsewhere, and the recursion's terminal arrival cost delivers the single scalar corner condition. Run in exact $\mathbb{Q}$, acceptance becomes a tolerance-free theorem at the $O(Nd^3)$ operation count of one LQ Riccati sweep, modulo the border-carrying extension of §3.

What exactness buys here, and what it does not. When an active keep-out guards more than one coordinate, the optimum need not be rational. Rational contact must be established; inverse design is one way to ensure it (Remark 2). Without a rational closing witness the deliverable is an exact rational *lower bound*. A rigorous lower bound is not unique to exact arithmetic: interval methods and verified Cholesky factorizations [19, 36] deliver one at fixed precision. This paper claims neither speed nor a tighter bound. What exact arithmetic delivers that no positive-width enclosure can is, first, *decidable equality*: closure $t^\star = J^\star$ and exact contact $f_j(x^\star) = 0$ are $\mathbb{Q}$-decidable, and no positive-width enclosure can certify an equality — it can bound a gap below ε, never establish that it is 0. Second, *auditability*: the certificate is a finite rational object that any independent implementation re-verifies bit-for-bit, provided it meets the implementation and data requirements of §8 — a correct implementation of the checks, exact rational arithmetic, unbounded integers, the transmitted data — with no tolerance to agree on and no dependence on platform rounding. The closing certificates of Theorem 1 (Appendix A) exercise the equality deliverable on instances whose rational optimum is designed in — the constructed families of Definition 3 among them; §6 states what survives without that design.

Delta over the published technical note. The published Podium technical note [29] establishes the exact S-procedure bracket and the rational $LDL^\top$ PSD test for *single* primitives; the mathematics there is classical and presented as such. This paper's delta is the *trajectory* case:

1. **An auditable bound with a structured check (§2, §3).** The general-case deliverable is an exact rational lower bound — weak duality needs no rationality of the optimum — carried by a certificate that is a finite rational object, re-verified bit-for-bit by any correct implementation meeting the three re-verifier conditions of §8. The arithmetic count follows from the band identity: the band-restricted PSD test *is* the LQ Riccati recursion where the block pivots stay positive, with the scalar band-$LDL^\top$ sweep carrying the same verdict

elsewhere, so testing $H(\lambda) \succeq 0$ costs $O(Nd^3)$ exact rational operations (Lemma 1, *proved*); full acceptance of $M(\lambda, t) \succeq 0$ extends this to the bordered sweep at the same $O(Nd^3)$ order — which is where the headline order is won or lost, since M has no bandwidth — and that extension is *proved* too (Lemma 2: border locality is inherited from the bandwidth of H, and the zero-pivot branch is exactly the range condition of the generalized Schur complement), while the checker itself is sign and zero tests. The protocol is sound but not complete (refusal asserts nothing; Definition 1).

2. **A machine-checked predicate and executable checkers (§3, §5).** The *acceptance predicate* — a nonnegative-weight decomposition exists iff the matrix is PSD — is *machine-checked* in Lean in both directions (the converse yields a rank-one sum of unfixed shape, not a square $L^\top DL$); that the shipped sweep computes such a factorization is not machine-checked (§3, [30]). The bordered sweep ships as a trusted checker in the Podium library, and the pipeline is *demonstrated* end-to-end on an inverse-designed CW-coupled stress instance (§5), every reported number reproduced by an executable exact-arithmetic receipt [30].

2 Problem and certificate protocol

Let $x = (x_1, \ldots, x_N)$ with each $x_k \in \mathbb{R}^d$ and total dimension $n = Nd$; the optimization is over $\mathbb{R}$, while the problem *data* and the *certificate* are rational. Consider

$$J^\star = \min_x \; f_0(x) \quad \text{s.t.} \quad f_j(x) \geq 0, \; j = 1, \ldots, m, \tag{1}$$

with a block-tridiagonal objective coupling only neighbours,

$$f_0(x) = \sum_{k=1}^{N-1} \|x_{k+1} - A_k x_k\|_{R_k}^2 + \sum_{k=1}^{N} (x_k - g_k)^\top W_k (x_k - g_k),$$

with $R_k \succ 0$ the stage-cost weight and $W_k \succeq 0$ the goal weight, and per-stage keep-outs

$$f_j(x) = \|\Pi_j x_{\sigma(j)} - c_j\|^2 - \rho_j^2, \qquad \rho_j > 0, \tag{2}$$

each involving only the single stage $\sigma(j) \in \{1, \ldots, N\}$; several keep-outs may guard the same stage. Here Π_j is a fixed coordinate projection selecting the *guarded* sub-block of the stage state: $\Pi_j = I$ is the full-stage keep-out ball, while a guidance keep-out typically guards the position sub-block of a position–velocity state (§5). The neighbour-only coupling confines every nonzero entry to a scalar band of half-width $b = 2d - 1$, and that is the band the sweep of §3 uses.

The certificate's shape can be read off the one-dimensional picture first. At the minimum of a parabola the tangent is horizontal: the derivative vanishes at the minimizer, and that first-order identity is an algebraic fact about the minimizer's own coordinates — a claimed minimizer is checked by evaluating a derivative, not by rerunning any solver. When the minimum sits on the boundary of the feasible set — here, on an active keep-out sphere — the gradient need not vanish. Under a qualification on the active gradients (linear independence of the active constraint gradients, LICQ — one of the hypotheses of Theorem 1, Appendix A, where it pins the multipliers uniquely), the gradient is balanced by the active constraints' gradients with nonnegative weights, and those weights are the multipliers λ of the KKT stationarity condition. The balance alone does not certify. With nonconvex keep-outs many points satisfy it: local minima, saddle points, contacts on the wrong branch of a sphere. The S-procedure certificate (λ, t) defined next is the object that closes the gap: acceptance upgrades "these coordinates balance" to the global statement $t \leq J^\star$,

and closure of the bracket (Definition 1) upgrades it again, to global optimality of the claimed point — checked in exact arithmetic throughout. Stationarity motivates closing certificates; a general lower-bound candidate need not satisfy stationarity at the feasible point supplied for the upper bound.

Write $f_0(x) = x^\top P_0 x + q_0^\top x + r_0$ with P_0 block-tridiagonal, and $f_j(x) = x^\top P_j x + q_j^\top x + r_j$. Then $P_j = \Pi_j^\top \Pi_j \succeq 0$ is supported on the single block $\sigma(j)$, and since Π_j is a coordinate projection, $\|\nabla f_j\| = 2\rho_j$ on the contact sphere. These are the structural properties of the keep-outs that the results below use; Lemma 3's (Appendix B) sharp form additionally takes $A_k = I$, $R_k = I$, $W_k = w_k I$ and $\Pi_j = I$. A multiplier vector $\lambda \in \mathbb{Q}^m_{\geq 0}$ and a value $t \in \mathbb{Q}$ certify the lower bound $t \leq J^\star$ when

$$M(\lambda, t) = \begin{bmatrix} H(\lambda) & \frac{1}{2}(q_0 - \sum_j \lambda_j q_j) \\ \frac{1}{2}(q_0 - \sum_j \lambda_j q_j)^\top & r_0 - \sum_j \lambda_j r_j - t \end{bmatrix} \succeq 0, \qquad H(\lambda) := P_0 - \textstyle\sum_j \lambda_j P_j,$$

by weak duality: $M \succeq 0$ makes $f_0 - \sum_j \lambda_j f_j - t$ globally nonnegative, so $t + \sum_j \lambda_j f_j$ is a global minorant of f_0. The converse fails: Example 1 (Appendix B) exhibits true lower bounds admitting no level-one certificate. Acceptance of a certificate (Definition 1) is therefore the sufficient direction.

By a generalized Schur complement, $M(\lambda, t) \succeq 0$ holds iff three conditions do:

(i) $H(\lambda) \succeq 0$;

(ii) the linear part $\frac{1}{2}(q_0 - \sum_j \lambda_j q_j)$ lies in range$(H(\lambda))$ — automatic when H is nonsingular, and when H is singular the sweep detects the condition as a consistent zero-pivot row;

(iii) the scalar corner condition $t \leq g(\lambda)$, where $g(\lambda)$ is the Lagrangian dual function value at λ.

Remark 1 (Curvature convention). Block tridiagonality of $H(\lambda)$ is forced by the class rather than assumed: every summand of f_0 touches only $\{x_k, x_{k+1}\}$ or x_k alone, each P_j sits in the single block $\sigma(j)$, and block-tridiagonal matrices form a linear subspace. The structure is load-bearing: the band sweep cannot see out-of-band entries, and it returns a wrong verdict on a symmetric matrix that lacks the structure. Under the convention $f_0 = x^\top P_0 x + q_0^\top x + r_0$ used here, with no $\frac{1}{2}$ factor, $H(\lambda)$ is half the Lagrangian Hessian: $\nabla^2_{xx}(f_0 - \sum_j \lambda_j f_j) = 2H(\lambda)$. We therefore call $H(\lambda)$ the *Lagrangian curvature matrix*. The factor is fixed by that convention and is load-bearing wherever H enters quantitatively: H, and not $2H$, is the curvature block of $M(\lambda, t)$, and the pivot bit-sizes §5 measures are those of H. What is invariant under a positive rescaling are the three verdict-level properties invoked below — semidefiniteness, weak diagonal dominance, and pivot signs — since the band pivots are homogeneous of degree one in H.

Definition 1 (Certificate protocol)**.** The *S-procedure certificate* is the pair (λ, t); its acceptance — $\lambda \geq 0$ and $M(\lambda, t) \succeq 0$, checked in exact arithmetic — proves $t \leq J^\star$. The *emitted certificate* is the object a prover transmits: the pair $(\lambda, \bar{x})$ of multipliers and a witness. The verifier checks $\lambda \geq 0$ and $f_j(\bar{x}) \geq 0$ for every j. It computes the upper bound $U = f_0(\bar{x})$ by exact evaluation. Independently, it assembles $H(\lambda)$, $\ell = q_0 - \sum_j \lambda_j q_j$ and $\gamma_0 = r_0 - \sum_j \lambda_j r_j$, and runs Lemma 2's bordered sweep with initial corner γ_0 (that is, at $t = 0$). If curvature or the range condition fails, it refuses. Otherwise the returned corner is the finite dual value

$$t = g(\lambda) = \inf_x \big(f_0(x) - \textstyle\sum_j \lambda_j f_j(x)\big),$$

and $M(\lambda, t) \succeq 0$ follows from the same sweep with zero terminal corner. A negative returned value is still a valid lower bound; this step does not require $M(\lambda, 0) \succeq 0$. The interval is $[g(\lambda), U]$. No primal stationarity test is required, and no proposed objective value is transmitted or trusted.

For a stationary witness, evaluating the Lagrangian at $\bar{x}$ is an optional shortcut: $H(\lambda) \succeq 0$ and $2H(\lambda)\bar{x} + \ell = 0$ give $g(\lambda) = \mathcal{L}_\lambda(\bar{x})$. The inverse-designed family drivers use this shortcut and verify the resulting $M(\lambda, t)$ explicitly. Section 4 uses the general arrival-value route. The two outcomes are asymmetric, in the standard sense that the protocol is *sound* but not complete. Acceptance carries the bound as a theorem: no tolerance exists through which a false certificate could pass. A candidate failing any check is *refused*, and a refusal asserts nothing — in particular, it does not prove the candidate suboptimal (Appendix B exhibits an instance for which no level-one certificate attains the true optimum). Here level one is the constant-multiplier S-procedure relaxation. Level two of the moment/SOS hierarchy allows a degree-four SOS remainder and degree-two SOS multipliers on each quadratic constraint [23]. Throughout, the verifier works in exact rational arithmetic, evaluating f_0 and the f_j and assembling $M(\lambda, t)$ from its own copy of the problem data: a certificate is meaningful only relative to agreed data. The interval $[t, f_0(\bar{x})]$ *brackets* $J^\star$, and the bracket *closes* when $t = f_0(\bar{x})$. Every size or cost claim below names which object it concerns; for claims about a certificate, that is the S-procedure certificate or the emitted certificate.

The translated-interval problem: a nonstationary feasible witness. For $\min x^2$ subject to $(x-3)^2 \geq 25$, take $\lambda = 1/4$ and $\bar{x} = -2$. The Lagrangian is $3x^2/4 + 3x/2 + 4$, whose minimizer -1 is infeasible. The general protocol nevertheless returns $[\frac{13}{4}, 4]$. Substituting the feasible witness into the Lagrangian instead gives 4, for which M has determinant $-9/16$. Thus a valid multiplier need not admit any feasible stationary witness. `check_revision.py` checks both matrices and the accepted bracket.

Standing assumptions. Two assumptions, A1 and A2, are referenced throughout.

Assumption 1 (A1: aggregate-PSD exactness)**.** There exist a feasible $x^\star$ and multipliers $\lambda^\star \geq 0$ satisfying the KKT conditions of (1) — stationarity of $f_0 - \sum_j \lambda_j^\star f_j$ at $x^\star$, and complementary slackness $\lambda_j^\star f_j(x^\star) = 0$ — such that the Lagrangian curvature matrix satisfies $H(\lambda^\star) \succeq 0$.

The quantifier is existential, and deliberately so: A1 asserts that such a pair exists, not that some given point is one. Nothing in the assumption asks the verifier to *decide* KKT status — what the verifier checks is $\lambda^\star \geq 0$, $M(\lambda^\star, t^\star) \succeq 0$ and feasibility, all decidable over $\mathbb{Q}$ — the PSD test by §3's sweeps, feasibility by exact rational evaluation of the f_j. In the constructed families of §5 the pair is supplied by the design, so A1 is there a property the construction arranges; on a given instance A1 is a hypothesis about that instance, and Appendix B gives a solver-free sufficient condition for its PSD clause: weak diagonal dominance of the assembled H, an entry-wise test with no eigencomputation and no knowledge of $x^\star$, which for the full-stage identity-coupled chain of Lemma 3 reduces to the per-stage load condition $w_k \geq \lambda_k$ — checkable there from the stage goal weights and multiplier loads alone; the existence of the KKT pair itself is not what that shortcut supplies. Neither reading requires certifying KKT status as part of the certificate.

Assumption 2 (A2: rational recoverable optimum)**.** A1's witness $x^\star$ is rational: the keep-out surfaces active at it, $\|\Pi_j x_{\sigma(j)} - c_j\| = \rho_j$, meet the stationarity variety — the zero set in x of $\nabla\big(f_0 - \sum_j \lambda_j^\star f_j\big)$ — at the rational point $x^\star$ itself.

A1 is the S-procedure exactness condition. Under it, global optimality of $x^\star$ becomes a *conclusion* (Theorem 1), not a hypothesis, and checking A1's PSD clause is a solver-free PSD test on $H(\lambda^\star)$ that needs no prior knowledge of optimality; it is implied by the weak-diagonal-dominance rule of Appendix B, which is proved there but is not exercised on the demonstrations of Section 5. A2

is the arithmetic condition that exact closure additionally requires; Remark 2 shows that rational data alone do not supply it.

Remark 2 (Rational contact is not automatic). A2 is arithmetic-geometric: rational data need not place an optimum at a rational point of the active sphere. Rationality must therefore be established; our closing test families ensure it by construction. Inverse design is not necessary. For every rational $0 < a < 1$, the directly specified problem $\min(x-a)^2+y^2$ subject to $x^2+y^2 \geq 1$ has unique optimum $(1,0)$ and multiplier $1-a$, with $H = aI \succ 0$. Indeed the distance to $(a,0)$ is at least $\sqrt{x^2+y^2}-a \geq 1-a$, with equality only there. No statement about generic frequency follows in either direction. Designing rational contact *via the standard constructions* of Definition 3(i) (Appendix A) — Pythagorean triples for $d = 2$, stereographic projection of rational parameters for general d — requires $\rho_j \in \mathbb{Q}$: they reach rational points only when ρ_j is rational. That is a limit of those constructions, not of rational contact itself: §5's grid-snapped family designs exact rational contact in with ρ_j irrational, by taking ρ_j^2 from the snapped point. At guard dimension 2 or 3 a rational non-square ρ_j^2 can leave the sphere with no rational point to design in at all: $x^2+y^2 = 3$ and $x^2+y^2+z^2 = 7$ have none. Dimension 4 or more always admits one, by Lagrange's four-square theorem. A keep-out guarding a single coordinate with rational radius $\rho_j \in \mathbb{Q}$ is the exception: contact there pins that coordinate to the rational value $c_j \pm \rho_j$ directly. What (2)'s data fix is only $\rho_j^2 \in \mathbb{Q}$: with ρ_j^2 a rational non-square, single-coordinate contact is always irrational, while a higher-dimensional sphere may still carry a rational point the constructions just named do not reach, as $(1,1)$ does on $x^2+y^2 = 2$. Tolerance-free closure thus *requires* an instance whose optimum is rational — A1 is needed besides (Example 1); §6 states what survives without A2.

2.1 Size measures

All data are rational, and every bit bound in this paper is stated against a height — in the sense of Definition 2 — of a tuple that the bound names. Three statements stand outside the scheme: the per-entry bit-size of the rationalized Clohessy–Wiltshire (CW) state-transition matrix in §5, stated against the denominator cap that produces it; the 20-bit-per-coordinate cap the fixed grid puts on the companion family's emitted denominators in the same section; and the imported 2^n-bit lower bound of [6], stated against the problem dimension.

Definition 2 (Height and bit-size)**.**

(a) For a finite tuple T of rationals, its *height* $\mathrm{ht}(T)$ is the least h such that T can be written over one common denominator, $T = (a_1/D, \ldots, a_r/D)$ with D a positive integer and the a_i integers, subject to $\lceil \log_2(D+1) \rceil \leq h$ and $\lceil \log_2(|a_i|+1) \rceil \leq h$. That least h is attained at the least common multiple of the entries' reduced denominators, which is the D the receipts compute. Clearing that denominator turns the tuple into integers of at most h bits — which is what Hadamard's and Edmonds' bounds require.

(b) A single rational p/q in lowest terms has bit-size $\mathrm{bit}(p/q) = \lceil \log_2(|p|+1) \rceil + \lceil \log_2(q+1) \rceil \leq 2\,\mathrm{ht}(p/q)$; that per-entry measure is what Table 3 reports for the emitted certificate.

(c) For a positive *integer* D we write $\mathrm{bitlen}(D) = \lceil \log_2(D+1) \rceil$ for its bit-length, the measure the receipts apply to denominators; read as a rational, $\mathrm{bit}(D) = \mathrm{bitlen}(D) + 1$.

Table 1 collects the named heights the bounds below are stated against; each is the height, in the sense of Definition 2, of a different tuple, and §5 measures all four.
These are different tuples and in general different numbers, and no ordering among L, L_{asm}, L_M and $\mathrm{ht}(x^\star)$ is forced, so each bound below names which height it uses. The corner $r_0 - \sum_j \lambda_j r_j - t$

Table 1: The named heights.

Height	tuple measured	where used
L (*per-stage design height*)	the design tuple Definition 3 places on a stage: that stage's placed coordinates, its active keep-out centres c_j, and its active multipliers $\lambda_j^\star$	Theorem 1's bit-size leg, stated against this tuple and no larger one
L_{asm} (assembled height)	the assembled coefficients P_0, q_0, r_0, together with the goals g_k that q_0 and r_0 are recovered from	Remark 4's (Appendix A) bounds on the certified value
L_M (bordered height)	the bordered matrix $M(\lambda, t)$ that the sweep eliminates: $L_M = \mathrm{ht}(M(\lambda, t))$	Remark 3's pivot bit-size bound
$\mathrm{ht}(x^\star)$	the placed coordinates taken over *all* stages at once	Remark 4's sharper first inequality

of the bordered matrix can cancel r_0's denominator against t's, and on the demonstration family of §5 it does. Nor is $\mathrm{ht}(x^\star)$ a renaming of L: L bounds a stage's design tuple one stage at a time, while $\mathrm{ht}(x^\star)$ is taken over the placed coordinates of all stages at once.

3 The certificate is a Riccati sweep

$H = H(\lambda)$ is symmetric block-tridiagonal, with $d \times d$ blocks H_{kk} and couplings $H_{k,k+1} = -A_k^\top R_k$. Its scalar bandwidth is at most $b = 2d - 1$. Equality holds whenever some coupling block has a nonzero $(1, d)$ entry, as it does on the family of §5; structured couplings such as $A_k = R_k = I$ instead sit at half-width exactly d for $N \geq 2$, strictly inside the band whenever $d \geq 2$. The sweep uses the conservative b in either case. Eliminating blocks in increasing order produces the pivots

$$S_1 = H_{11}, \qquad S_k = H_{kk} - H_{k,k-1}\, S_{k-1}^{-1}\, H_{k-1,k} \quad (\text{while } S_{k-1} \succ 0), \tag{3}$$

and, in the positive-definite interior, $H \succ 0$ iff every $S_k \succ 0$; the general semidefinite verdict, including singular boundary pivots, is delivered by the scalar band-$LDL^\top$ of Lemma 1 with its zero-pivot handling.

Equation (3) is the discrete *Riccati / block-Schur* recursion of an LQ problem: S_k is the accumulated (arrival-cost / information-form) curvature matrix after eliminating $x_1, \ldots, x_{k-1}$ — a Schur complement in H, so it inherits the same no-$\frac{1}{2}$ convention as $H(\lambda)$ (Remark 1). That the block-banded KKT factorization equals the Riccati recursion is classical [41, 35]; here it is run in exact $\mathbb{Q}$.

The full certificate is $M(\lambda, t) \succeq 0$, that is, the three conditions of §2. Carrying the affine term $\frac{1}{2}(q_0 - \sum_j \lambda_j q_j)$ through the same sweep makes the terminal corner pivot $g(\lambda) - t$; equivalently, run the sweep at $t = 0$ and read off the arrival cost $g(\lambda)$. All three conditions are thus certified in one tolerance-free rational sweep of $O(Nd^3)$ operations. That sweep is an extension of Lemma 1; its *soundness* is *proved* as Lemma 2, and its operation count is a bound on the shipped sweep's loops.

Lemma 1 (Band-$LDL^\top$ soundness)**.** *For symmetric H with bandwidth b, the $LDL^\top$ that eliminates only within the band yields the same pivots — hence the same PSD verdict: refuse at a negative*

pivot, and at a zero pivot refuse unless its current row is zero — as the dense test, in $O(n(b+1)^2) = O(Nd^3)$ exact rational operations.

Band preservation is proved below; the content of the lemma is the exact-rational, zero-pivot-handling PSD *verdict*.

Proof. Eliminating pivot k updates $a_{ij} -= (a_{ik}/a_{kk})a_{kj}$ for $i, j > k$. With bandwidth b, $a_{ik} \neq 0 \Rightarrow i \leq k + b$ and $a_{kj} \neq 0 \Rightarrow j \leq k + b$, so only entries with $i, j \in (k, k + b]$ change and the bandwidth is preserved; by induction all nonzero updates lie in the band, so eliminating only there is exact. At a zero pivot, positive semidefiniteness requires $a_{kj} = 0$ for all $j > k$; bandwidth makes this automatic for $j > k + b$, so checking $j \in (k, k + b]$ suffices. Each pivot touches an $\leq b \times b$ window and performs a pivot check, giving $O(n(b + 1)^2)$, including a diagonal matrix ($b = 0$). □

$M(\lambda, t)$ has no bandwidth: its last row and column carry the generally dense $\frac{1}{2}(q_0 - \sum_j \lambda_j q_j)$, so Lemma 1 applied to M gives $O(n^3)$, not $O(Nd^3)$. The bordered sweep instead carries the border as a *separate* column, at one rank-one corner update per pivot. Lemma 2 proves that the border-carrying sweep is sound and that its operation count is $O(Nd^3)$.

Lemma 2 (Bordered sweep soundness)**.** *Let $H \in \mathbb{Q}^{n\times n}$ be symmetric with bandwidth b, let $\ell \in \mathbb{Q}^n$, $\gamma \in \mathbb{Q}$, and set*

$$M \;=\; \begin{pmatrix} H & \frac{1}{2}\ell \\ \frac{1}{2}\ell^\top & \gamma \end{pmatrix} \in \mathbb{Q}^{(n+1)\times(n+1)}.$$

Run the band $LDL^\top$ of Lemma 1 on H while carrying $c := \frac{1}{2}\ell$ as one extra column: at a pivot with $a_{kk} > 0$ update $c_i -= (a_{ik}/a_{kk})\, c_k$ for $i \in (k, k + b]$ and $\gamma -= c_k^2/a_{kk}$; at $a_{kk} < 0$ refuse; at $a_{kk} = 0$ refuse unless $a_{kj} = 0$ for $j \in (k, k + b]$ and $c_k = 0$, and otherwise skip the index. Then the sweep refuses exactly when $H \not\succeq 0$ or $\frac{1}{2}\ell \notin \mathrm{range}(H)$; when it does not refuse it returns $\gamma - \frac{1}{2}\ell^\top y$ for any y with $Hy = \frac{1}{2}\ell$ (the value does not depend on which y), and $M \succeq 0$ iff that value is ≥ 0. The sweep costs $O(n(b + 1)^2) = O(Nd^3)$ exact rational operations.

Proof. (0) The dense criterion. For symmetric A, natural-order symmetric elimination decides $A \succeq 0$ without pivoting: if $a_{kk} < 0$ then $A \not\succeq 0$; if $a_{kk} = 0$ then $A \succeq 0$ forces $a_{kj} = 0$ for all j, since the principal minor on $\{k, j\}$ is $-a_{kj}^2 \geq 0$, and the index may then be deleted; if $a_{kk} > 0$ then $A \succeq 0$ iff its Schur complement with respect to a_{kk} is.

(1) The sweep is that elimination on M. Order the indices of M as $1, \ldots, n$ (the H block) and then $n + 1$ (the border). Since $M_{i,n+1} = c_i$ and $M_{n+1,n+1} = \gamma$, eliminating pivot $k \leq n$ is exactly: the H-block update of Lemma 1, the border update $c_i -= (a_{ik}/a_{kk})c_k$, and the corner update $\gamma -= c_k^2/a_{kk}$. So the sweep performs (0) on M; what remains for that identification is that its restricted index ranges lose nothing, (2)–(3), and that the separately-carried border's zero-pivot test is exactly (0)'s row condition, (4); after which (5) identifies the accumulated corner, (6) reads the verdict off that corner and the two refusal branches, and (7) bounds the cost.

(2) H-block locality. By Lemma 1, $a_{ik} \neq 0 \Rightarrow i \leq k+b$, $a_{kj} \neq 0 \Rightarrow j \leq k+b$, and the bandwidth is preserved, so restricting the H-block update to $i, j \in (k, k + b]$ is exact.

(3) Border locality. The border column is dense, but the multiplier a_{ik}/a_{kk} vanishes unless $i \leq k + b$ by (2). Hence c_i changes only for $i \in (k, k + b]$: at most b entries per pivot. Locality of the border update is inherited from the bandwidth of H and requires no structure in ℓ.

(4) The zero-pivot branch. If $a_{kk} = 0$, (0) requires the whole current row k of M to vanish: $a_{kj} = 0$ for $j \in (k, n]$ *and* $c_k = 0$. By (2) the first condition holds automatically for $j > k + b$, so testing $j \in (k, k + b]$ suffices; the second is not implied by bandwidth — the border is dense — and is tested separately. This is exactly the generalized Schur complement's range condition

$\frac{1}{2}\ell \in \mathrm{range}(H)$, index by index: each elimination acts as a row operation on the augmented system $(H \mid \frac{1}{2}\ell)$ together with the matching column operation on H, so after any prefix of pivots the current state is $(EHE^\top, E\frac{1}{2}\ell)$ with E invertible; since $\mathrm{range}(EHE^\top) = E\,\mathrm{range}(H)$ ($E^\top$ being invertible), $Hy = \frac{1}{2}\ell$ is solvable iff the current system is, solutions corresponding by $y = E^\top w$ — a zero current row k with $c_k \neq 0$ therefore certifies $\frac{1}{2}\ell \notin \mathrm{range}(H)$, while if the sweep reaches the end — in particular every zero pivot passed the test — (5) exhibits a solution. So when $H \succeq 0$ — the case in which no other branch can refuse first, and in which a zero pivot has its band row zero automatically by (0) — this branch refuses exactly on range failure; without that hypothesis a refusal here may instead be a nonzero band row, which by (0) already witnesses $H \not\succeq 0$. When it passes, row and column k of the current matrix are zero and the index is deleted; the corner is untouched, correctly, since $c_k = 0$ contributes nothing.

(5) The accumulated corner. If the sweep reaches the end, its pivots and multipliers assemble $H = LDL^\top$ exactly, with L unit lower-triangular and $D = \mathrm{diag}(a_{kk})$ — a skipped index contributes an identity column of L and a zero of D, exactly, since its current row and column are zero. Each c_k is frozen at pivot time after precisely the eliminations L^{-1} encodes, so the vector of pivot-time values is $z = L^{-1}\frac{1}{2}\ell$. By (4)'s test $z_k = 0$ at every skipped index, so setting $u_k := z_k/a_{kk}$ at the pivots used and $u_k := 0$ at the skipped ones gives $Du = z$, whence $y := L^{-\top}u$ satisfies $Hy = LDL^\top y = LDu = Lz = \frac{1}{2}\ell$ — the solution (4) promised, which also makes $\frac{1}{2}\ell \in \mathrm{range}(H)$ whenever the sweep does not refuse. For any y with $Hy = \frac{1}{2}\ell$, setting $u = L^\top y$ gives $Du = z$, and $(\frac{1}{2}\ell)^\top y = (LDu)^\top y = u^\top Du = \sum_k z_k^2/a_{kk}$ over the pivots used — the total the sweep subtracts from the corner — skipped indices contributing nothing since $z_k = 0$ there by (4)'s test. The value does not depend on which y: two solutions differ by an element of $\ker H$, on which $\frac{1}{2}\ell$ has no component, because $\frac{1}{2}\ell \in \mathrm{range}(H) = \ker(H)^\perp$ for symmetric H. Equivalently, writing H^+ for the Moore–Penrose pseudoinverse, the returned value is $\gamma - \frac{1}{4}\ell^\top H^+ \ell$.

(6) The verdict. By the generalized Schur complement criterion, $M \succeq 0$ iff $H \succeq 0$, $\frac{1}{2}\ell \in \mathrm{range}(H)$, and $\gamma - (\frac{1}{2}\ell)^\top y \geq 0$. The sweep refuses exactly on the first two failures, each refusal certifying the failure listed against it — a negative band pivot or a zero pivot with a nonzero band row certifies the first, a zero pivot whose band row is zero but with $c_k \neq 0$ the second, and the first refusal reached wins when both hold — and otherwise returns the third quantity for its sign to be tested.

(7) Cost. Each of the n pivots performs $O(b^2)$ band updates by (2), $O(b)$ border updates by (3), and $O(1)$ corner work, giving $O(n(b+1)^2) = O(Nd^3)$ with $b = 2d-1$, $n = Nd$. □

Two independent checks accompany the proof. Its verdict claim (6) and its value claim (5) are exercised against the dense M-test and against a separate exact solve of $Hy = \frac{1}{2}\ell$ on adversarial instances the ordinary tests do not reach — singular H with the zero pivot placed first, mid-sweep and last, indefinite H, ℓ exactly in $\mathrm{range}(H)$, ℓ purely on $\ker(H)$, ℓ mixing range and kernel components, and $\ell = 0$ (360 cases, 80 with indefinite H; no mismatch on either claim — 360/360 verdicts and 160/160 value checks, the remaining instances refusing, with no corner value to compare); and the receipt's own transcription of the sweep is pinned pivot by pivot against the dense $LDL^\top$ of M at the shortest horizons of §5's family. The $O(Nd^3)$ count of (7) is a bound on the shipped loops; no test measures the bordered sweep's operation count, it is the band sweep's count that is measured against its loop bound. The companion note catalogues the pinning receipts [30]. The *dense criterion* the lemma proves the sweep against is additionally *machine-checked* in Lean at general order over any linear ordered field: steps (4), (5) and (6) — acceptance data implies PSD; PSD forces the kernel-orthogonal border and the corner bound; the corner value is independent of the solution chosen, with the equivalence packaged under solvability of $Hy = \frac{1}{2}\ell$ and the range condition carried in its kernel-orthogonality form — in `BorderedCriterion.lean`. The implication

from that orthogonality to solvability remains on paper. Step (0)'s elimination criterion was already machine-checked order-generically by the acceptance-predicate corpus ([30]). The locality steps (2) and (3), the cost bound (7), and the algorithm-to-criterion bridge also remain on paper — that the sweep's recurrences compute the criterion's data, i.e. the congruence and factorization arguments inside steps (1), (4) and (5) above. The border's position is neither proved here nor mechanized: the mechanized statements carry the border at the head of the matrix where M carries it last, and the two layouts are identified by a simultaneous row/column permutation under which PSD is invariant — a standard fact, used and not proved.

Both tests ship as trusted checkers in the Podium library. The checkers work in exact rational arithmetic, refuse floats, and reject malformed input — non-rational entries, mismatched shapes, asymmetric blocks — before any verdict. One entry point is the band Hessian test. The other is the corner-carrying bordered sweep: one band $LDL^\top$ carrying the affine column, with the singular-H range condition detected as a consistent zero-pivot row, delivering the full $M(\lambda, t) \succeq 0$ verdict in $O(Nd^3)$ exact rational operations. Both entry points materialize the $n \times n$ matrix before eliminating, so their memory — and hence their total work — is at least quadratic in n; the $O(Nd^3)$ of Lemma 1 counts the sweep's rational *operations*, which the assembly does not change. A fixed-seed test suite pins both checkers against their dense counterparts — the band-H sweep on the assembled H, the bordered sweep on the assembled M, the latter being the dense M-test the library's reference checker performs — over randomized instances plus dedicated singular-H and zero-pivot cases. The companion note names the checkers, the pinning suite, and the library commit that carries them [30].

Remark 3 (Coarse effectiveness). The sweep's pivots are ratios of consecutive leading minors of the bordered matrix. The zero-pivot branch does not disturb that identity. By Lemma 2(4) a skipped index carries an identically zero row and column of the current matrix, and deleting a zero row and column changes no minor on the remaining indices; the elimination that continues is therefore the ordinary one on the retained index set S, whose k-th pivot is the ratio of the two consecutive leading minors of M restricted to the first k and first $k-1$ retained indices. So the identity holds verbatim with "leading minor" read on S rather than on $\{1, \ldots, n+1\}$. Clearing the common denominator that $L_M = \mathrm{ht}\big(M(\lambda, t)\big)$ supplies, their bit-sizes are

$$O\big((n+1)(L_M + \log(n+1))\big)$$

by Hadamard's inequality and the classical exact-elimination argument of Edmonds [14], and this PSD check runs in time polynomial in n and L_M. Evaluating the independently supplied upper witness also depends on its rational height. The order $n+1$ is the worst case over the sweep, attained at the *bordered* $(n+1) \times (n+1)$ matrix itself: only the terminal pivot's minors reach the border, the interior pivots' minors being leading minors of the H-block alone, so bounding every pivot at the bordered order and the bordered height is conservative. The implied constant is not 1, and the factor is stated rather than absorbed: after clearing the common denominator D that L_M supplies, a pivot is a ratio of two consecutive cleared minors *divided by* D — the integer-minor ratio is D times the pivot — so its cancellation-free denominator is D times a minor, not a minor. Both factors still fit the display: the denominator's minor has order at most n, one below the bordered order $n+1$ the display is instantiated at, and $\mathrm{bitlen}(D) \le L_M$ fits inside that spare order of Hadamard headroom. Numerator and denominator are therefore each bounded by the displayed expression and the pivot's bit-size by their sum — at most twice the display. The display is a bound on one minor; the pivot costs two, the D riding in the order slack just named. The height that governs here is the bordered matrix's, not a per-stage height, and the difference is not cosmetic. Assembly multiplies denominators across the coupling, so on the demonstration family of §5 the bordered

height dwarfs the per-stage design height, and no per-stage height bounds even the *entries* of the matrix the sweep eliminates, let alone its pivots. §5 measures those pivots inside the displayed expression formed with L_M at every horizon, and outside the one formed with the design height L. Sharp growth laws are model-sensitive; §5 reports measured growth. For the general protocol's computation at $t = 0$, instantiate this bound with $\mathrm{ht}(M(\lambda, 0))$. The closing-family measurements instead test $M(\lambda, t^\star)$ directly and use its height; the two matrices need not have the same height. Neither is a bound on the cost of finding a multiplier.

Remark 3's displayed expression takes both the bordered height L_M and the order $n{+}1$ of the bordered matrix because what the sweep costs is bounded through the two together, and through neither alone. This is witnessed by a pair of admissible families (Definition 3, Appendix A), distinct from Table 3's rows, each closing and each constructed by an executable receipt catalogued in the companion note [30]. The first family has bordered matrix entries all in $\{-1, 0, 1, 2, 3\}$, $x^\star = 0$, unit keep-outs contacted at $c_k = -1$, and multipliers alternating 1 and 0. It holds $L_M = 2$ at every horizon while its interior pivots are the consecutive-continuant ratios of $\mathrm{tridiag}(-1, [1, 3, 2, 3, 2, \dots], -1)$, its final diagonal entry one smaller because the endpoint row has degree one, whose two-step transfer matrix has spectral radius $2 + \sqrt{3}$, so those continuants grow geometrically and the largest interior pivot's bit-size passes any fixed budget — 12 bits at $n{+}1 = 9$ and 483 at $n{+}1 = 257$ — so no function of L_M alone bounds the cost. The second family fixes the order at $n{+}1 = 9$ while its multipliers' denominators grow: at $\lambda = 1/q$ the first interior pivot is exactly $(2q - 1)/q$, whose bit-size rises with q without bound, so L_M and the largest interior pivot rise together — L_M from 10 to 262 bits and that pivot from 85 to 4116 — and no function of the order alone bounds the cost either.

The acceptance predicate — a nonnegative-weight decomposition exists iff the matrix is PSD — is machine-checked in Lean in both directions (the converse yields a rank-one sum of unfixed shape, not a square $L^\top D L$); that the shipped sweep computes such a factorization is not, and the companion note records what is machine-checked and under which hypotheses [30].

4 Observations: numerical proposals on fixed problems

These are three selected examples of the general protocol, not a benchmark population or a coverage claim. The rational objectives, constraints, starting points and rationalization rule are fixed before running a solver. They are: the translated-interval problem of §2; the disk problem $\min(x - 1/3)^2 + (y - 2/5)^2$ subject to $x^2 + y^2 \geq 1$, whose unique optimum $(5, 6)/\sqrt{61}$ is irrational; and Example 1's two-stage problem.

SLSQP proposes points feasible to solver tolerance from the specified starts, and Clarabel solves the level-one dual SDP through CVXPY. Their values are untrusted. The receipt rounds each primal point to a grid of step 1/10000, checks the neighboring grid points exactly, and keeps the feasible point with least objective among those candidates. It replaces each numerical multiplier by a rational with denominator at most 10000, then applies Definition 1. A fixed schedule also tries common multiplier scales $1 - 10^{-6}$, $1 - 10^{-4}$, 99/100 and 0; every attempt is retained. Each run has a distinct receipt with its status and any proposal errors. A new run invalidates the preceding table; a table is emitted only after all three examples finish with accepting attempts. Failed runs retain their records and do not supply a table. Table 2 reports the first accepting attempt; the unscaled rational proposal is accepted for all three examples in the recorded completed run. This rule neither guarantees a feasible rational point nor guarantees a tight bound on other instances.

The disk row shows that a bound remains available without a rational optimum. The two-stage row has an irreducible level-one gap, proved in Example 1; a smaller rationalization error cannot

Table 2: Selected numerical-proposal examples. Displayed endpoints round the exact rational bracket outward to three decimal places. The JSON receipt retains the exact fractions, numerical proposals, all attempts and software versions.

Fixed problem	lower endpoint	upper endpoint
Translated interval	4.000	4.000
Irrational disk contact	0.229	0.230
Two-stage relaxation gap	12.705	16.308

remove it. These runs exercise the numerical-to-exact interface. They do not establish performance, success rates or scalability on independently sourced trajectory problems. The following constructed families instead test exact closure and intermediate arithmetic against known answers.

5 Observations: inverse-designed CW-coupled instances

We instantiate (1) as a *certification stress instance with Clohessy–Wiltshire structure* — not a flyable maneuver. Consecutive stages are coupled through the planar CW state-transition matrix Φ (mean motion $n_0 = 10^{-3}\,\mathrm{rad/s}$ and sampling phase $\tau = n_0\,\Delta t = 1/2$, so $\Delta t = 500\,\mathrm{s}$) via the soft-dynamics *full-state penalty* $\|x_{k+1} - \Phi x_k\|_R^2$: there is no control variable and no thrust model, the keep-out radius cycles $\{1/2, 1, 3/2\}$ per stage, and the instance is *constructed by inverse design* (each stage's guarded contact placed on a rational keep-out-sphere point, goals recovered from stationarity, multipliers chosen so A1 holds — a *benign* instance in Appendix B's vocabulary). The stage state is $x_k = (x, y, v_x, v_y) \in \mathbb{Q}^4$ in the LVLH frame, with x radial (outward-positive) and y along-track — the convention that fixes the sign of the no-drift condition below — and the keep-out guards its *position* sub-block: Π_k in (2) selects (x, y), giving $\|\Pi_k x_k\|^2 - \rho_k^2 \geq 0$ about the target at the LVLH origin, with the two velocity coordinates placed rationally by the design. It demonstrates the certification pipeline on a rational-optimum instance whose *coupling matrix* is the CW state-transition matrix, rationalized entrywise (below); the *trajectory* is not CW-consistent and is not meant to be. A second CW family in this section follows the rationalized dynamics to within a fixed grid step — a residual of 10^{-6} order rather than the 0 of exact propagation — at a measured cost in certificate size; the two share the same assembly, multipliers and goal weights, and differ in how the keep-out radii are fixed — prescribed radii met by design, versus radii taken from the trajectory — and in where the designed optimum is placed.

A stage is one sampled state: the horizon variable N counts state samples taken $\Delta t = 500\,\mathrm{s}$ apart, and one sampling step advances the reference orbit's phase by $\tau = 1/2\,\mathrm{rad}$, so about 12.6 steps span one orbit. The N stage states are joined by $N-1$ coupling residuals, so a horizon spans $(N-1)\,\Delta t$. At the top of the demonstration window, $N = 24$, that is $11500\,\mathrm{s}$: about $3.2\,\mathrm{h}$, or 1.8 orbits in the model's units. The same horizon carries $n = Nd = 96$ scalar variables at $d = 4$ per stage. The translation to time is specific to the two CW families of §5: the shared-state families of Table 3 are abstract chains with no sampling interval.

The design places the optimum where stationarity requires it: at N=24 the per-step dynamics residual $\|x_{k+1}^\star - \Phi x_k^\star\|$ runs 134.8–244.1 against states of norm 0.559–1.572, and the goals recovered from stationarity reach $\|g_k\| \approx 1.01 \times 10^5$. Units are nominal — positions in metres, velocities in metres per second — so $R = I_4$ and $W_k = w_k I_4$ are not dimensionally homogeneous weights and the norms just quoted mix the two, consistent with the instance not being a flyable maneuver. So this does not discover an optimum for an operational CW problem, and the goal term is a

goal-attraction penalty, not a tracking objective. The CW state-transition matrix is rationalized entrywise to denominators at most 10^5, giving per-entry bit-size at most 42. The rationalization breaks the symplectic structure: after rounding, $\det \Phi \approx 1 + 1.19 \times 10^{-8}$ — a deviation computed exactly over $\mathbb{Q}$, not a floating-point artifact — against $\det \Phi = 1$ for the CW flow. The certificate is therefore a theorem about the *rational model only*: it certifies optimality and keep-out satisfaction for the rationalized dynamics, not for the continuous CW flow. The rounding moves each entry by at most 2.8×10^{-10} in absolute value and 2.0×10^{-7} relative; it is the 10^5 denominator cap, not the 42-bit entry size, that sets those figures.

Finite phase schedules. In `rpod_cw.py`, the contact direction rotates with period 4, radii and multipliers have period 3, and weights and velocities have period 6. Their common period is 12; stationarity therefore gives periodic interior goals, with finitely many endpoint phases. `check_cw_height.py` checks the resulting constant-term increment.

A dynamically consistent companion family, and what it costs. The design just described places its optimum at *fixed rational height*, which is what makes the emitted certificate's per-entry size independent of the horizon; whether a fixed-height design whose trajectory satisfies the dynamics *exactly* exists is not settled here. Making the trajectory dynamically consistent instead — propagating $x^\star_{k+1} = \Phi x^\star_k$ exactly — drives the residual to 0 and the goals below the state norm, but a free-drift state is $\Phi^k x_0$. Powers of this rational matrix can accumulate denominator factors. The receipt `freedrift_exact_check.py` propagates without snapping and places the radii on those states, so the dynamics residual vanishes by construction. It checks stationarity against the assembled Lagrangian and the positive diagonal remainder $w_k > \lambda_k \geq 0$. The pinned bordered-PSD checker accepts the resulting lower bound and refuses a raised one; the feasible-upper checker returns the same objective value. Over $N = 2, \ldots, 24$, the value's measured bit-size grows from 274 to 7698; the largest entry bit-size of the emitted pair $(\lambda^\star, x^\star)$ grows from 78 to 3814. These are finite measurements of different objects, not asymptotic growth proofs.

A stepwise snapped companion gives a different tradeoff. Propagate and snap *stepwise*, $x^\star_{k+1} := \mathrm{snap}(\Phi x^\star_k)$ with every entry of snap's output rounded to the nearest multiple of 10^{-6} — a *fixed* grid, so all stages share one denominator and no least common multiple can compound across the horizon; the residual below is $\|x^\star_{k+1} - \Phi x^\star_k\|$, each step measured against the snapped predecessor — and take $\rho_k^2 := \|\Pi_k x^\star_k\|^2$, which is rational as a sum of rational squares and puts the guarded sub-block *exactly* on its sphere by construction rather than by designing radii to be met. Initialising at $x_0 = (1, 0, 0, -2n_0)$ — a grid point with $\rho_0 = 1$ meeting the CW no-drift condition $\dot{y}_0 = -2n_0 x_0$ — nulls the secular along-track mode of the exact flow; each snap then perturbs the drift invariant $\dot{y} + 2n_0 x$ by up to the grid step, so the constructed orbit is near-bounded over the demonstrated horizons (state norm ≤ 2.03 at $N \leq 24$, against ≈ 2 for the unsnapped propagation — from that x_0 the exact flow's state-norm supremum is $\sqrt{4 + n_0^2}$, its position part alone peaking at 2; both figures are properties of this initial state, not of the no-drift condition, which leaves the ellipse's size and along-track centre free), not bounded structurally. This is row 4 of Table 3 (driver `rpod_cw_freedrift.py`), at all 23 horizons closing, contacting exactly, and agreeing across the three routes — the bundle's own block-Riccati sweep standing in for the trusted band-H checker on this row:

$$\text{residual} \leq 7.1 \times 10^{-7} \text{ against states of norm} \leq 2.03, \qquad \|g_k\| \leq 1.64,$$

against 244.1 and 1.01×10^5 for the inverse-designed family — a trajectory whose per-step dynamics residual sits at the 10^{-6} snap grid, and goal norms below the states they attract.

The cost is visible: emitted bits rise from 6–8 to 40–41 (the 10^{-6} grid is ≈ 20 bits per coordinate) and value bits from 931–1089 to 990–1116. Of the emitted figure, only the denominator half is pinned structurally — the fixed grid caps it at 20 bits per coordinate at any horizon — while the numerator tracks the state norm, which the snap-excited drift grows beyond the demonstrated range: extending the generator, emitted bits rise $\approx \log N$, from 41 at N=24 to 42 at N=100 and 49 at N=8000, with state norm 2.03 $\to$ 313 (receipt `freedrift_longhorizon_check.py`). Over the demonstrated horizons the emitted figure is a measurement (40–41), like the value figure, not a structural cap. The height L of the family's per-stage tuple measures 22–23 bits, per `freedrift_value_check.py`, and Definition 2's $\mathrm{bit}(\cdot) \le 2L$ then covers the 40–41 *identically* rather than by measurement — the emitted entries are members of the very tuple L is taken over, as on row 3 — so that covering carries no horizon-independence: it is not Theorem 1(ii)'s bound, which is stated only for constructed families and does not apply to this one (see the table caption), and L is itself measured here rather than placed. The value figure is *measured* over the demonstrated horizons, and Remark 4's ceiling is verified on this family too: its divisibility law and all three size inequalities hold at every horizon (receipt `freedrift_value_check.py`), with $\mathrm{bitlen}(\mathrm{den}(t^\star))$ running 496–557 against the per-horizon bound $L_{\mathrm{asm}} + 2L$ of 560–621 bits. Both sides reach their observed plateaus near $N \approx 13$ (L_{asm} at 575, the denominator at 557) and stay within two bits of the plateau thereafter — 556 and, for $\mathrm{bit}(t^\star)$, 1114 at N=24 — saturation observed over the demonstrated horizons. Remark 4(1) proves boundedness of the denominator, not these plateau values. The numerator depends on both the number of stage terms and their magnitudes; the snapped trajectory's state norm is measured rather than uniformly bounded over all horizons. The Riccati pivot bit-sizes are *identical* between the two families at every horizon — necessarily so: $H(\lambda) = P_0 - \sum_k \lambda_k P_k$ depends on Φ, R, the weights and the multipliers, and on none of the designed point, the goals or the radii, so no choice of where to place the optimum can move the PSD structure the certificate rests on. Everything in §3 and Remark 3 that is stated against H therefore transfers unchanged. That remark's *bound* does not: it is instantiated at the bordered height $L_M = \mathrm{ht}(M(\lambda, t))$, which moves with the design — 277–284 bits on the inverse-designed family against 303–313 on this one — as do M's corner entry and the certified value.

What this does *not* repair: the rationalized Φ still breaks symplecticity, so both families remain theorems about the rational model only; and $R = I_4$ still mixes position and velocity units.

Everything is checked in $\mathbb{Q}$ by the three routes of §3 — the band-H test, the corner-carrying bordered sweep on the full M, and the dense M-test — which agree on every row of Table 3, with the bundle's own block-Riccati sweep standing in for the band-H test on row 4. At every one of the 23 horizons $N = 2, \ldots, 24$ the bracket closes for both CW families, so weak duality already forces $g(\lambda^\star) = t^\star$ and a terminal corner pivot of exactly 0; as a sanity check the bordered sweep refuses $t^\star + 10^{-6}$ at every horizon, and the guarded position sub-blocks of the optima lie exactly on the keep-out spheres.

The emitted pair is small wherever its entries are placed at a fixed height — 18 bits or fewer in rows 1–3, though only rows 2–3 are placed in that sense: row 1's maximum falls on a multiplier recovered from stationarity, not on a placed entry — and 40–41 in the grid-snapped row 4, whose entries are supplied by the grid rather than placed by the definition's constructions and where the grid pins only the denominator half at 20 bits, the numerator — which tracks the state norm — being of comparable size over the demonstrated horizons — while the recomputed CW *value* ($\sim 10^3$ bits) comes from the inverse design, not from the raw data. The goal term $r_0 = \sum_k w_k \|g_k\|^2$ *squares* the denominators recovered from stationarity: $\mathrm{den}(t^\star)$ and $\mathrm{den}(r_0)$ coincide at every horizon from N=6 to N=24 (both 519 bits at N=24), and $\mathrm{den}(t^\star)$ divides $\mathrm{den}(r_0)$ at 22 of the 23 horizons — the exception is N=5, where the two differ by a factor 6 the other way. Here the goal term is the dominant driver, not the sole one — a property of this inverse design and not of Definition 3. Per

Table 3: Exact-rational trajectory certificates (all verdicts exact over $\mathbb{Q}$). "emitted bits" is the worst-entry bit-size of the emitted $(\lambda^\star, x^\star)$; "value bits" is the size of the certified value $t^\star$ the verifier recomputes (Remark 4). The size column carries both units: N is the number of stages, d the per-stage state dimension, and $n = Nd$ the number of scalar variables. The regime column places each family in the vocabulary of Appendix B: *benign* means A1 holds at the optimum. The refusal row's instance is itself benign — its unconstrained minimizer is a feasible exact KKT pair with $H(0) \succ 0$ (receipt `refusal_row_a1_check.py`) — what that row exhibits is soundness: its *emitted certificate* designs $\lambda_k > w_k$ on a full-stage identity-coupled instance, the sweep measures $H(\lambda^\star) \not\succeq 0$, and the checker refuses, issuing no false certificate. (The design is what makes the instance a refusal test; the non-PSD verdict is measured by the sweep rather than inferred from $\lambda_k > w_k$ — an inference this instance does license, being Lemma 3's free-endpoint boundary family with a uniform excess, where WDD is necessary as well as sufficient, but which is unavailable in general because WDD is only sufficient for $H \succeq 0$.) Instance-level A1 failure is Example 1's, whose certified gap rules out every PSD KKT pair. Rows 2 and 3 are constructed rational-sphere families (Definition 3). Row 4 is *not* one: it meets clauses (ii)–(iv) and clause (i)'s activity and rationality conditions — every keep-out active at a rational contact point of positive radius, LICQ, rational multipliers with A1, goals from stationarity — but not clause (i)'s placement route: the snapped trajectory supplies the contact points directly, with $\rho_j^2 \in \mathbb{Q}$ always and ρ_j itself irrational at every stage past the initial one (stage 0 sits at the no-drift state, a grid point with $\rho_0 = 1$; receipt `freedrift_longhorizon_check.py`), points the clause's constructions cannot reach (Remark 2). Like row 1 — the scalar shared-state family, whose odd stages carry a degenerate radius-zero keep-out ($\rho_k = 0$, outside (2)'s $\rho_j > 0$) with solved rather than placed coordinates, inert since $\lambda_k = 0$ there and deleting those constraints leaves $x^\star, \lambda^\star, t^\star$ and the certificate unchanged — row 4 is benign and covered by Theorem 1's closure leg but not by its $O(L)$ bit-size bound, which is stated only for constructed families; both rows' bit columns are therefore empirical measurements. Rows 3 and 4 are the two CW designs of §5 — the same dynamics and the same Hessian, differing in how the keep-out radii are fixed and in where the designed optimum is placed — and their ranges are over all 23 horizons $N = 2, \ldots, 24$; rows 1 and 2 are over the horizons the measurement receipt samples, up to the max size shown.

Family	regime	max size	closes	emitted bits	value bits
Scalar shared-state	benign	N=129 (n=129)	yes	13–18	13–26
Vector shared-state ($d = 1..4$)	benign	n=Nd=96	yes	3–4	15–32
CW-coupled keep-out	benign	N=24 (n=96)	yes	6–8	931–1089
CW free-drift keep-out	benign	N=24 (n=96)	yes	40–41	990–1116
Non-PSD certificate refused	benign	N=4 (n=8)	no (refuses)	—	—

Remark 4 it is the *denominator* that is bounded in N: $\mathrm{den}(t^\star)$ stays inside 450–526 bits at every horizon out to $N = 6143$. The size keeps the remark's residual $\log N$: it runs 931–1089 bits over the 23 horizons $N = 2, \ldots, 24$, non-monotone, drifting past that range only beyond the measured window (1090 bits first at N=46, 1097 by N=6143). The optional multiplier reconstruction (Theorem 1) shows the same split between the design and assembled heights. The solve's *operands* stay at the design height: 5–6 bits against $L = 6$ on this family, and 259 bits against $L = 258$ on the inverted family noted inside Definition 3's closing paragraph. They stay there because stationarity makes the objective gradient the multiplier combination of the active keep-out gradients, which are design-tuple data. The *formation* of that gradient from the assembled coefficients is bounded through the assembled height together with the height of the placed coordinates and active centres, whichever dominates being a property of the instance (as Theorem 1's proof states): on this family the assembled side dominates — its two terms run 242–276 bits against $L = 6$ and cancel to the design height — while on the inverted family the design side does, the formed gradient staying at 259 bits against $L_{\mathrm{asm}} = 3$–8.

The heights, measured. The four separate heights of §2.1 are "in general different numbers" on this family in the strongest sense: at every one of the 23 horizons its four tuples carry four pairwise distinct heights, in the strict order $\mathrm{ht}(x^\star) < L < L_M < L_{\mathrm{asm}}$; Table 4 reports the measured ranges.

Table 4: The four heights of §2.1, measured on the CW-coupled demonstration family over all 23 horizons $N = 2, \ldots, 24$; they are pairwise distinct at every horizon, in the strict order $\mathrm{ht}(x^\star) < L < L_M < L_{\mathrm{asm}}$.

Height	measured	what it governs here
$\mathrm{ht}(x^\star)$	5 bits at every horizon	Remark 4's sharper first inequality
L	6 bits, attained at every horizon (per-stage values 3, 5 or 6)	Theorem 1's $O(L)$ leg; Table 3's emitted 6–8 bits
L_M	277–284 bits	Remark 3's pivot expression
L_{asm}	483–563 bits	Remark 4's bounds on the certified value; carries the 519-bit $\mathrm{den}(r_0)$

The per-stage *design* tuple of Definition 3(iii) — a stage's placed coordinates, its keep-out centre (the LVLH origin here) and its multiplier — has height at most $L = 6$ bits at every stage of every one of the 23 horizons, the maximum attained at every one of them. Table 3's emitted 6–8 bits sit next to that figure by construction rather than by measurement: the emitted pair is part of the very tuple L is taken over, so $\mathrm{bit}(\cdot) \le 2L$ (Definition 2) holds identically, here and on any constructed family. This is a bound relative to a design tuple that already contains the emitted entries; it proves neither compression nor efficient recovery on an independently specified instance. There are $Nd + m$ emitted entries, so a per-entry bound is not a total-size bound. Over the displayed horizons neither figure grows past N=3 — L is 6 bits at every horizon, and the emitted size is 8 bits at every $N \ge 3$, 6 at N=2. The assembled coefficients (P_0, q_0, r_0, g) have height $L_{\mathrm{asm}} = 483$–563 bits, consistent with Definition 3(iv), which makes r_0 derived data of the assembled height rather than of L: the 519-bit $\mathrm{den}(r_0)$ above is carried by L_{asm} and could be carried by neither the design height nor L_M. And the assembled bordered matrix $M(\lambda^\star, t^\star)$ that the sweep eliminates has height $L_M = 277$–284 bits, the smaller of the two assembled figures because its corner $r_0 - \sum_j \lambda_j^\star r_j - t^\star$ cancels those two large denominators against each other.

Instantiating Remark 3's expression at the bordered order $n{+}1$ gives $(n{+}1)\big(L_M + \lceil \log_2(n{+}1) \rceil\big)$, running 2529–28227 bits over the horizons. The measured *scalar* pivots of the corner-carrying sweep — the object Remark 3 bounds, up to that remark's factor of two — run from 777 bits at $N{=}2$ to 17893 at $N{=}24$ and stay inside that expression at all 23 horizons. (A second, close series appears below for the largest *entry* of the $d \times d$ block pivots S_k; it is a different measure of the same elimination, not a discrepancy.) Because the bracket closes, the terminal corner pivot $g(\lambda^\star) - t^\star$ is exactly 0 at every horizon, so the figure reported is the largest *interior* pivot of the corner-carrying sweep. The same expression formed with the design height L runs only 90–1261 bits and is exceeded at all 23 horizons. The pair of measurements $L_M = 277$–284 bits against $L = 6$ already says that on this family no per-stage height bounds even the *entries* of the bordered matrix, let alone its pivots. Here it is the bordered height, and not a per-stage height, that Remark 3's expression must be instantiated with.

That ordering is a property of the CW assembly and not of Definition 3, which forces neither direction: an exact receipt exhibits an admissible family, closing on all three checkers, on which $L_M \le 8$ bits stands against a design height $L = 257$–258, so there the per-stage design height bounds every entry of the bordered matrix [30]. Remark 4's three inequalities hold at all 23 horizons as well — the compact $\mathrm{bitlen}(\mathrm{den}(t^\star)) \le L_{\mathrm{asm}} + 2L$, the value bound $\mathrm{bit}(t^\star) \le 2(L_{\mathrm{asm}} + 2L) + \lceil \log_2 N \rceil + \lceil \log_2(3d^2{+}d{+}1) \rceil$, and the sharper first form $\mathrm{bitlen}(\mathrm{den}(t^\star)) \le L_{\mathrm{asm}} + 2\,\mathrm{ht}(x^\star)$ — as does the divisibility step behind them. The largest *entry* of the $d \times d$ block pivots S_k of (3) grows measurably linearly in the horizon — 780 bits at $N{=}2$ to 17889 at $N{=}24$. The scalar pivots quoted above are exactly the interior pivots inside those blocks, so what separates the two series is the entry-versus-pivot measure, not a different elimination. Over all 23 horizons they land within 11 bits of one another (median gap 4, worst case 11 at $N{=}19$, both signs), per `pivot_gap_allhorizons.py`. Neither figure is a correction of the other, and this is reported as data; growth laws are future work.

Reproducibility. Each row of Table 3 is constructed by a named driver in exact rational arithmetic — the four closing rows certified, the refusal row refused — and each driver verifies its emitted certificate through the exact dense M-test. The refusal row's *emitted pair* takes $\lambda_k > w_k$ at $N{=}4$, $d{=}2$; its instance is benign (see the caption). Measurement receipts recompute both bit columns at the sizes Table 3 headlines, re-verify the first three closing rows through all three trusted routes (the dense reference M-test, the band-H sweep, and the bordered-M sweep) — the fourth carrying its own receipt, which asserts closure, exact contact, and agreement between the dense bracket verdict, the Riccati sweep and the trusted bordered-M sweep at all 23 horizons — and recompute the CW-coupled family's denominator statistics at all 23 horizons; a shipped test suite re-runs the denominator-statistics receipt and re-checks both bit columns at the headline sizes. The trusted checkers refuse floats, and the CW state-transition matrix alone is synthesized in floating point and rationalized before any certification, as above. The revision package contains `supplementary-scripts-v26.zip`, with the existing family receipts, the new protocol and proposal receipts, a pinned checker snapshot and setup instructions. Its SHA-256 is `f31aa499f4c61347b8aaf42c39f7b9adc0932e5c07c1e8cf6917da199ec1f65a`. The proposal table uses `numerical-proposals.json`, run `7afe3301618e4e05a3cd9375698803b3`, with driver SHA-256 `f97728d635a908177d5754e13a7419c7b5d1518abf21de59bd2a2db39a2a4ffa`. The historical run did not record its Python interpreter version; `validation-environment.json` identifies the later exact check. The checkers originate in the Podium repository [31], described in [32]. The companion note catalogues the receipts and distinguishes the formal predicate from the executable implementation [30].

6 Scope and limitations

Rationality is needed for emitted closure, not for a lower bound. If no global minimizer is rational, no rational feasible witness can close the emitted bracket. Even then a rational multiplier with PSD curvature and a consistent border yields a finite exact lower bound. The multiplier and feasible upper-bound witness are checked independently; neither is presumed to be a KKT witness. Rational contact can fail as in Remark 2 and the disk row of Table 2. A keep-out guarding a single coordinate with rational radius instead pins the active coordinate to a rational value, leaving the remaining coordinates a rational linear solve when every other guard active there likewise guards a single coordinate with rational radius. Quantifying the resulting bracket (rates in the rationalization budget, behaviour at the relaxation-gap floor) is outside this paper's scope.

Verification bit-complexity is measured and coarsely bounded, not sharply bounded. This paper *measures* verification cost — the linear pivot growth of §5 — and bounds it coarsely (Remark 3); sharp growth laws and their model sensitivity are outside scope. That verification cost can grow along the horizon while each emitted certificate entry stays bounded is visible in the demonstration of §5.

The demonstration family's mildness is tied to its shared transition matrix. The CW-coupled family couples every stage through one shared rationalized transition matrix, computed for a circular reference orbit. An eccentric-reference variant of the same construction — the Yamanaka–Ankersen transition matrix, a different matrix at every stage — reproduces the CW pivot sequence bit for bit at $e = 0$. At the probed eccentricities $e \in \{0.1, 0.3, 0.5\}$, rationalized to the same realized fidelity, its sweep pivots grow at 1037–1049 bits per stage (the three eccentricities' measured slopes) against the CW family's ≈ 778; the excess does not track eccentricity magnitude — the slope at $e = 10^{-4}$ already exceeds the slope at $e = 0.1$ — and it is not attributed here, since the shipped comparison moves the per-stage transition matrix and the per-entry rationalization budget together. On the CW family the certified value's denominator is bounded in N by the finite phase argument in Remark 4; its measured band is 450–526 bits over the demonstrated horizons (§5). The value's bit-size has the stated logarithmic bound and the measured band in that section; on the variant at $e = 0.1$ the value's bit-size grows at ≈ 1000 bits per stage, so the compactness of the recomputed value does not carry over. An executable receipt catalogued in the companion note pins these figures [30].

The per-stage Hadamard/Cramer reconstruction bound is not included. A per-stage Hadamard/Cramer bound on the optional multiplier reconstruction is weaker than the $2L$ bound the design order gives, a consequence of the design order rather than a route to it, and not needed for either leg of Theorem 1.

Bienstock realizability inside the class is not established. Quadratic optimization at large admits 2^n-bit optima [6]; whether such a gadget can be realised as an instance of (1) is not settled either way. That is why Theorem 1 is stated conditionally, and why this paper does not claim conditioning is unavoidable.

Sample feasibility does not certify interpolation safety. The keep-outs constrain the N sampled states. For example, $f_0 = (x_2 + x_1)^2 + (x_1 - 1)^2 + (x_2 + 1)^2$ with $x_1^2, x_2^2 \geq 1$ has a closing certificate at $(1, -1)$ with $\lambda = 0$, objective zero and zero coupling residual for $A = -1$.

Linear interpolation nevertheless passes through the obstacle center. This limitation is separate from model rationalization and from allowing a nonzero dynamics residual. Continuous curve constraints, as in [16], require an additional argument.

A limited hard-dynamics interpretation. For $x_{k+1} = A_k x_k + B_k u_k$ with rational invertible square B_k and unconstrained u_k, and rational $R_{u,k} \succ 0$, eliminating the controls converts $u_k^\top R_{u,k} u_k$ exactly into the coupling penalty with $R_k = B_k^{-\top} R_{u,k} B_k^{-1} \succ 0$. The state and control formulations are then equivalent when their remaining costs and constraints agree. Underactuated maps, input bounds, or fixed-state equalities generally leave additional constraints; their treatment is outside this formulation. Wright and Rao–Wright–Rawlings [41, 35] treat the general equality-constrained factorization in floating point.

Level one need not close. Under A1, multiple keep-outs per stage already close (Theorem 1); when a stage's keep-out geometry itself violates A1 — several exclusion balls on one stage, two quadratic constraints where the S-lemma wants one, the sign-flipped analogue of the two-*inclusion*-constraint CDT subproblem [8, 2] — level-one certification can have a positive gap. Higher-order relaxations are possible, but this paper does not establish that level two suffices for every such instance.

7 Related work and positioning

Exact-rational certificates for optimization. The direct antecedent of tolerance-free certificates for optimization outputs is the exact rational LP/MIP line: VIPR's machine-verifiable exact certificates for integer-programming results [9], exact rational MIP solving in SCIP [15], and the exact LP solver QSopt_ex [4]; the bit-size control of exact elimination goes back to Edmonds [14]. Certificates measured by their verification complexity are the program of [13]. Ours carries that line's exact-certificate discipline to a structured nonconvex quadratic class, with the banded-KKT/Riccati structure doing the work.

Trajectory SDP relaxations and tightness. Sparse moment–SOS for trajectory optimization [21] and perception [43] is the closest numerical program: floating-point, tolerance-based certificates — STROM reports certified suboptimality below 1% for most of its five trajectory problems, near 1% for the hardest (car back-in), and Yang and Carlone report a sparse relaxation that is *empirically exact*, deeming a rounded solution globally optimal at a relative suboptimality below 10^{-3} — obtained from level-two (sparse second-order) relaxations (for STROM, the order that closes gaps its own ablation shows level one leaves open; for Yang and Carlone, the minimum order their degree-three truncated-least-squares objective admits), with tightness observed rather than proved. The two lines are complementary: exact-arithmetic acceptance on benign, rational-optimum (constructed) instances versus broad numerical coverage. On the same constraint geometry as Appendix B, Graesdal et al. [16] give necessary *and* sufficient tightness conditions for a *single*-sphere-obstacle trajectory SDP relaxation; tightness in their own many-obstacle experiments is measured empirically, not obtained from that theorem. Ours is only sufficient and holds with any number of keep-outs per stage — of which LICQ leaves at most d active on each — but is solver-free and checkable in exact arithmetic along the sweep, and their single-obstacle conditions delimit what a sharp benign boundary could look like. In the adjacent domain of spacecraft conjunction avoidance, Vega et al. [38] solve collision-avoidance exclusion QCQPs via a Shor relaxation

observed empirically tight — on a different formulation, with control variables and conjunction-derived constraints, so not a measurement of A1 on our class. We share no benchmark with any of these, so this paragraph is positioning, not comparison. A general tightness framework for QCQP SDP relaxations [39], graph-structural exactness conditions [37], exactness on forest structures [5], and local-to-global exactness from clique-wise sub-SDPs [22] position A1 within a broader theory (a path is a tree, but our concave keep-outs need not satisfy their hypotheses); chordal conversion and PSD completion [44, 17] address the structural SDP side. Lossless convexification [1] exactly convexifies nonconvex *control*-set constraints via the maximum principle — complementary in mechanism to exact certificates for nonconvex *state* constraints.

Riccati-based semidefinite MPC. TinySDP [28] solves a lifted MPC problem with stage-wise obstacle exclusions using Riccati–ADMM and cached steady-state quantities. Its a posteriori safety test compares a lifted clearance margin with the trace gap between the lifted second moment and the physical state product. The resulting identity certifies geometric clearance at the checked samples. This is a close structural predecessor: it uses Riccati computations inside a numerical SDP solver. Here the supplied multiplier is verified by an exact rational bordered sweep to produce an objective lower bound. These are different certificates, and no solver-speed or bound-quality comparison with TinySDP is made.

Rational recovery and exact SDP. Exact rational SOS/PSD certificate recovery is due to [33, 20], with RealCertify the standard tool [25]: its multivariate route computes an approximate decomposition of a perturbed input with an *arbitrary-precision* SDP solver (SDPA-GMP) and recovers an exact rational one symbolically, while its univariate `univsos1` uses no SDP at all — a recursive construction over $\mathbb{Q}$ from root isolation and quadratic under-approximations. Exact algorithms for LMIs and the SPECTRA library are the exact-SDP lineage [18]. Bit-size bounds for exact SOS certificates — univariate [27], multivariate [26, 11] — do not exploit band structure, an orthogonal axis to Theorem 1; machine-checkable SOS certification is developed in [24], the Coq-interfaced companion to the template method of [3].

Verified solvers and a-posteriori numerics. A complementary line proves the *solver* correct before it runs (credible autocoding and verified convex-optimization codes [40, 12, 10]); Roux et al. [36] validate solver output a posteriori via a floating-point Cholesky with proved rounding-error bounds — rigorous but tolerance-based. The a-priori and a-posteriori guarantees compose with ours rather than compete. On *why exactness*: a rigorous inequality certificate need not be exact. Interval arithmetic or a verified Cholesky [19, 36] certifies $H(\lambda) \succ 0$ with a positive margin, hence a rigorous lower bound, at fixed precision. A positive-margin criterion cannot reach the semidefinite boundary, and the closing certificates of Theorem 1 sit exactly there: at closure the corner Schur complement is exactly 0, and singular H is the case §2 handles through the consistent zero-pivot row. Fixed precision reaches the boundary only in a narrow case: the operations witnessing the boundary contact must be exact in the working precision, so that the boundary quantity is enclosed with zero width. Pivots away from the boundary may round, their positive-width enclosures absorbed by their positive margins. Even in that case a Cholesky does not certify the boundary, because it stops at an exactly computed zero pivot: on the singular H with rows $(1, -1)$ and $(-1, 1)$, where an $LDL^\top$ and the symmetric eigensolver are exact, LAPACK's `dpotrf` declines the factorization, reporting the order-2 leading minor not positive definite. The rationalized CW instance of §5 is not of that boundary-case kind at all: the denominators of its rationalized STM's nonintegral entries are not powers of two. Exact receipts pin each of these boundary facts; the companion note

catalogues them [30]. Exact rational arithmetic is likewise essential for the *equality* deliverables, decidable closure $t^\star = J^\star$ and exact contact $f_j(x^\star) = 0$. Both are $\mathbb{Q}$-decidable and auditable, but no finite-precision enclosure of positive width can establish them: such an enclosure can bound a gap below ε, never certify it is 0. A zero-width enclosure can, and fixed precision delivers one only in the exact-witness case named above. We claim exactness for auditability and decidable equality, not as the only route to a rigorous bound.

8 Conclusion

For a supplied rational multiplier, the band and bordered sweeps decide curvature and range consistency and compute the exact Lagrangian lower bound. A separately checked feasible trajectory supplies the upper bound. The general protocol requires neither a rational optimum nor stationarity at that trajectory. Under A1, A2 and LICQ, the corresponding rational KKT pair closes the bracket; the constructed families make that outcome available as an exact test oracle.

Auditability requires a correct implementation of the stated checks and three re-verifier conditions: exact rational arithmetic, unbounded integers, and the transmitted problem data. Rederiving a rounded transition matrix instead of reading its transmitted rationals can change the certified instance. The Lean results formalize the mathematical acceptance predicate, not the Python implementation. The examples test particular executions; they do not establish operational success rates or scalability. Current implementations use dense storage, and the constraints apply only at samples.

Appendix A Exact closure on designed instances

Role in this paper. The closure theorem below is conditional on a rational KKT witness (A2). Rational data do not ensure one (Remark 2); the closing families used here supply it by construction. The closure leg also applies to rational witnesses obtained by other routes under its stated hypotheses. Their value is as exact test oracles — instances whose answer is known perfectly, so any deviation by a correct re-verifier holding the same data is a defect. `attack3_scope.py` compares an IEEE-754 transcription with the exact checker on such a closing instance. Without the closure hypotheses, a valid multiplier still supplies the exact lower bound of §3.

Definition 3 (Constructed rational-sphere family)**.** A *constructed rational-sphere family* is a family of instances of (1) carrying a distinguished rational point $x^\star$, the *designed point* — Theorem 1, not this definition, is what makes it the optimum — in which the following four conditions hold.

(i) Every keep-out is *active* at $x^\star$, with radius $\rho_j > 0$, and each *guarded* contact point $\Pi_j x^\star_{\sigma(j)}$ is placed at a rational point of its keep-out sphere via a Pythagorean or stereographic parametrization — or, for a guard of dimension one, by pinning the coordinate to $c_j \pm \rho_j$, Remark 2's exception — the remaining stage coordinates being placed rationally as well, so that each stage's placed coordinates have height at most L.

(ii) At every stage the active constraint gradients $\nabla f_j(x^\star)$ are linearly independent. For one keep-out per stage this is automatic, since $\rho_j > 0$; for several keep-outs per stage it is a condition on the contact geometry. Since those gradients are supported on the one block, it caps the number of active keep-outs on a stage at d.

(iii) The multipliers $\lambda^\star_j \geq 0$ are chosen rationally so that, together with the stage's placed coordinates and keep-out centres, they form a per-stage design tuple of height at most L, and so that aggregate-PSD (A1) holds.

(iv) Every W_k is invertible, and the goals g_k are then recovered from stationarity, so the g_k — and with them q_0 and r_0 — are *derived* data, carried by the assembled height L_{asm} rather than by L.

The definition forces no ordering between L and L_{asm}: on the demonstration family (§5) L_{asm} is by far the larger, while an executable receipt exhibits an *inverted* family, admissible under this definition, on which it is the smaller [30]. No coordinate is left inactive to be solved for, so the emitted pair $(\lambda^\star, x^\star)$ is rational, of per-stage height at most L and hence of per-entry bit-size at most $2L$ (Definition 2), by construction.

Theorem 1.

(i) Closure. *Under Assumptions 1 and 2, with linear independence of the active constraint gradients (LICQ) at $x^\star$, the exact-rational bracket closes: there exist rational $(\lambda^\star \geq 0,\ t^\star = J^\star,\ x^\star)$ with $M(\lambda^\star, t^\star) \succeq 0$, $f_j(x^\star) \geq 0$ and $t^\star = f_0(x^\star) = J^\star$ — an accepting certificate in the sense of Definition 1 — with any number of keep-outs per stage, of which LICQ leaves at most d active on each.*

(ii) Emitted size and recovery. *For a constructed rational-sphere family (Definition 3) the emitted certificate $(\lambda^\star, x^\star)$ additionally has* per-entry *bit-size at most $2L$, independent of N and of d, and a verifier holding the instance data and the transmitted witness $x^\star$ recovers the same $\lambda^\star$ within that bound, so the transmitted $\lambda^\star$ need not be trusted.*

The bound in (ii) is on the *size of the recovered object*, not on the cost of recovering it, and the two are genuinely different here. The recovery's solve keeps its *operands* at the design height on every family measured, but *forming* the objective gradient from the assembled coefficients passes through terms of 242–276 bits on the demonstration family before they cancel back to design height (§5 measures both, and exhibits an inverted family on which they do not cancel) — while what the solve returns is the placed multiplier, whose entries the design pins at height L. Reading (ii) as a verification-cost statement would therefore overstate it: verification cost is governed by Remark 3 and the bordered height L_M, not by L.

Proof. Closure. By Assumption 1, $x^\star$ is a KKT point with multipliers $\lambda^\star \geq 0$; for a constructed family they are supplied outright, by Definition 3(iii)–(iv). A constraint qualification is not what produces them, since the theorem does not assume $x^\star$ is a local minimizer: optimality is the conclusion. By A1 the Lagrangian $\mathcal{L}(x) = f_0(x) - \sum_j \lambda_j^\star f_j(x)$ is convex and $x^\star$ is a stationary point, hence a global minimiser: $\min_x \mathcal{L} = \mathcal{L}(x^\star) = f_0(x^\star)$, the last step by complementary slackness. Thus $M(\lambda^\star, t^\star) \succeq 0$ at $t^\star = f_0(x^\star)$, and weak duality gives $t^\star \leq J^\star$, while feasibility gives $J^\star \leq f_0(x^\star) = t^\star$. Hence $t^\star = J^\star = f_0(x^\star)$.[1] Nothing in this argument uses one keep-out per stage: with several active keep-outs on a stage the multipliers enter $\mathcal{L}$ additively and complementary slackness holds term by term. Two exact receipts with two active keep-outs on one stage accompany this [30]: one with disjoint-coordinate contacts, closing at $t^\star = 169/320$, and one whose contacts' gradients share their whole support, closing at $t^\star = 5773/3600$.

Rationality of the multipliers. The data and $x^\star$ are rational, so stationarity $\nabla f_0(x^\star) = \sum_j \lambda_j \nabla f_j(x^\star)$ together with complementary slackness ($\lambda_j = 0$ at every inactive j) is a rational linear system in the active multipliers, and under LICQ — item (ii) of Definition 3 for the constructed families — it has a unique solution, which is therefore rational.

Bit-size of the emitted pair. A2 alone does not bound the height: for quadratic optimization at large the KKT system is polynomial, and Bienstock-type gadgets satisfy A2 with 2^n-bit optima [6]. In a constructed family the design order of Definition 3 settles the emitted pair directly: $x^\star$ by (i) and $\lambda^\star$ by (iii) are *placed* rationally at per-stage height at most L, and only then are the goals recovered by (iv). The placement leaves no coordinate to solve for: every keep-out being active, the guarded coordinates sit on their keep-out sphere and the rest are placed freely, with no inactive coordinate left over. So $(\lambda^\star, x^\star)$ has per-entry bit-size $O(L)$, independent of N.

What the verifier reconstructs. What needs an argument is that $\lambda^\star$ is exactly what a verifier holding the instance data and the transmitted witness $x^\star$ reconstructs. Fix a stage s, let A_s be the keep-outs active there and $m_s = |A_s|$; their gradients $\nabla f_j(x^\star) = 2\Pi_j^\top(\Pi_j x_s^\star - c_j)$ are supported on block s, hence live in $\mathbb{Q}^d$, and have Euclidean norm $2\rho_j > 0$, and by (ii) they are linearly independent, so $m_s \leq d$. Write $G_s \in \mathbb{Q}^{d \times m_s}$ for their block-s restrictions and $v_s = [\nabla f_0(x^\star)]_s$. Stationarity reads $G_s \lambda^\star_{A_s} = v_s$ with G_s of full column rank, so the normal equations

$$\big(G_s^\top G_s\big)\lambda^\star_{A_s} = G_s^\top v_s$$

have $\lambda^\star_{A_s}$ as their unique solution, and the verifier can reconstruct the emitted multipliers rather than trust them. Definition 1 does not require this solve: it checks the transmitted multiplier independently. What this optional solve reconstructs is the *placed* $\lambda^\star$ itself, so Definition 3(iii) together with bit $\leq 2\,\mathrm{ht}$ (Definition 2) bounds it by $\mathrm{bit}(\lambda_j^\star) \leq 2L$ — the design order, not the solve, is what makes the recovered object small. Nothing here asks the active gradients on a stage to have disjoint supports: the shared-support receipt above exhibits a Definition-3-admissible stage

[1] The mechanized counterparts of the weak-duality step and the attainment leg, and the hypotheses those theorems take, are itemized in the companion note [30].

on which they share every coordinate, and checks that the normal-equation solve recovers $\lambda^\star$ there. What that bound does *not* cover is the cost of the same solve run from the assembled instance alone. That solve's *operands* stay at the design height on every family measured, and not by accident: stationarity makes the objective gradient $[\nabla f_0(x^\star)]_s$ the multiplier combination of the active gradients $2\Pi_j^\top(\Pi_j x_s^\star - c_j)$, which are design-tuple data. What carries the assembled height is the *formation*: a verifier holding only the assembled coefficients and the transmitted witness builds $[\nabla f_0(x^\star)]_s$ as $2(P_0 x^\star)_s + q_0^{(s)}$. That formation is bounded through the assembled height together with the height of the placed coordinates and active centres, not through L alone, and which of the two dominates it is a property of the instance; §5 reports the measured figures. Acceptance recomputes $t^\star$ and runs the exact $LDL^\top$; the size of those intermediates is a *verification* cost (Remark 3), not a property of the emitted pair. □

Remark 4 (Size of the certified value). Write $\mathrm{den}(\cdot)$ for the lcm of the denominators of a tuple of rationals. At a *closing* certificate — where the recomputed dual value equals the feasible objective $f_0(x^\star)$ — the certified value $t^\star = f_0(x^\star)$ is a sum of N stage terms, the k-th the quadratic form in $x^\star$ carried by the *assembled* stage coefficients $(P_0^{(k)}, q_0^{(k)}, r_0^{(k)})$: block row k of P_0, block k of q_0, and a share $r_0^{(k)}$ of the constant, block-tridiagonality making that row reach only stages $k-1$, k and $k+1$. These value-size bounds are for closure: a general $g(\lambda)$ also involves the rational elimination of $H(\lambda)$ and can acquire denominators from its minors. Its computation is governed by Remark 3 at $M(\lambda, 0)$, not by evaluating the objective at the independent upper-bound witness. Any splitting of the constant with $\sum_k r_0^{(k)} = r_0$ and $\mathrm{den}(r_0^{(k)}) \mid \mathrm{den}(r_0)$ serves — for instance carrying all of r_0 on one stage — and the receipt takes the conservative $\mathrm{den}(r_0)$ at every stage. In four steps:

(1) Each summand's denominator divides $\mathrm{den}(P_0^{(k)}, q_0^{(k)}, r_0^{(k)}) \cdot \mathrm{den}(x^\star)^2$, the form being quadratic in the witness, and a finite sum's denominator divides the lcm of the summands', so

$$\mathrm{den}(t^\star) \;\Big|\; \mathrm{lcm}_{k \le N}\, \mathrm{den}\big(P_0^{(k)}, q_0^{(k)}, r_0^{(k)}\big) \cdot \mathrm{den}(x^\star)^2;$$

hence $\mathrm{den}(t^\star)$ is bounded in N whenever the coefficient denominators and the denominators of all placed coordinates come from bounded sets. Whether they do is a property of the family, and for the inverse-designed CW family it follows from the finite set of stage phases (§5, paragraph "Finite phase schedules"). The fixed-grid row 4 of Table 3 also has bounded coefficient and coordinate denominators: the grid, shared Φ, and periodic rational weights and multipliers fix their possible factors. Hence its value denominator is bounded in N as well. The observed plateau of value and assembled heights over the demonstrated horizons is a separate measurement. Bounded denominators do not bound numerators: the snapped family's numerator also depends on state magnitudes, for which no uniform bound over all horizons is established here.

(2) Stated against the heights of §2.1 — the *coefficient* height L_{asm}, not the bordered L_M, since it is the coefficient denominators that appear above — the right-hand side divides the common denominator that L_{asm} supplies times $\mathrm{den}(x^\star)^2$, so $\mathrm{bitlen}(\mathrm{den}(t^\star)) \le L_{\mathrm{asm}} + 2\,\mathrm{ht}(x^\star)$. Here $\mathrm{ht}(x^\star)$ is taken over all stages at once, whereas L bounds a stage's design tuple one stage at a time, so the more compact $\mathrm{bitlen}(\mathrm{den}(t^\star)) \le L_{\mathrm{asm}} + 2L$ additionally asks for a common representation of *all* placed coordinates of height at most L, that is, $\mathrm{ht}(x^\star) \le L$. This holds for the inverse-designed CW family of §5.

(3) The numerator adds only the $\log N$ of the sum over stages and the log of the number of monomials one block row contributes — at most $3d^2+d+1$, since block-tridiagonality leaves

block row k at most $3d$ nonzero columns across its d rows, plus q_0's block of d and the constant — giving $\mathrm{bit}(t^\star) \le 2(L_{\mathrm{asm}}+2L)+\lceil\log_2 N\rceil+\lceil\log_2(3d^2+d+1)\rceil = O(L_{\mathrm{asm}}+L+\log N)$ for fixed d, under (2)'s shared-denominator and height condition on the placed coordinates. For the inverse-designed CW family, the stage data, interior goals and endpoint goals range over finite sets; the global constant $r_0 = \sum_k w_k\|g_k\|^2$ is a sum of N nonnegative terms. Its denominator is bounded but its numerator grows. In particular, the interior goals are periodic with period 12, and fixed endpoint phases give $r_0(14+12k) = r_0(14) + k\Delta$ with $\Delta > 0$. Hence L_{asm} is unbounded, with $L_{\mathrm{asm}} = O(\log N)$, while L stays bounded. The value-size conclusion remains $O(\log N)$.

(4) The governing quantity is *not* the lcm of the raw data denominators, and cannot be: the assembled coefficients are products and quotients of the data — $A_k^\top R_k A_k$ multiplies two denominators, and recovering g_k from stationarity (Definition 3(iv)) divides by W_k, so a data *numerator* can become a coefficient denominator — after which the goal term squares them. On the CW family of §5 the raw-data lcm is 130 bits while $\mathrm{den}(t^\star)$ is 450–526 bits; the displayed divisibility holds at every one of the 23 horizons $N = 2, \ldots, 24$, and $\mathrm{den}(t^\star)$ divides the raw-data lcm at none of them.

§5 checks all three inequalities, and the divisibility step behind them, at every horizon of the demonstration family.

Why Remark 4 is stated against the heights it names, and why its bit bound carries a horizon term, is settled by admissible closing families, each constructed by an executable receipt [30]. The certified value's *denominator* is bounded through L_{asm} together with the height of the placed coordinates, but its *bit-size* additionally needs the horizon: an admissible closing family holds $L_{\mathrm{asm}} = 2$ and $\mathrm{ht}(x^\star) = 1$ at every horizon while $t^\star = N$ exactly. Nor is either quantity bounded through L alone: integer goal weights of growing magnitude take $\mathrm{den}(t^\star)$ unbounded at fixed $L = 1$, since goal recovery divides by W_k and the weight's numerator lands in the value's denominator.

No unconditional polynomial bit bound exists: quadratic optimization *at large* can force 2^n-bit exact solutions [6], and small-rational-solution existence is strongly NP-hard. What we *can* support is Theorem 1's guarantee for the constructed families, and the reason it is supportable is the design order of Definition 3: $(\lambda^\star, x^\star)$ is placed at per-stage height $O(L)$ before any datum is derived from it, so no solve stands between the design tuple and the emitted pair.

Appendix B Weak diagonal dominance: a checkable entry to A1

Role in this paper. Aggregate-PSD (A1) is the closure theorem's key hypothesis, and a conditional theorem is usable only with a checkable hypothesis. Weak diagonal dominance is that entry for A1's PSD clause: solver-free, proved, preserved along the sweep. It is an architectural device, not a frequency claim — how often real instances satisfy it is unmeasured, and the demonstration of §5 verifies A1 by the general sweep rather than by this shortcut. We call an instance *benign* when aggregate-PSD (A1) holds at its optimum — the S-procedure is then exact and the bracket can close. This is a property of the instance, not of a solver; this section gives a solver-free condition, checkable stage-by-stage along the sweep, that is sufficient for A1's PSD clause.

Lemma 3 (Solver-free benign criterion)**.** *Call H* weakly diagonally dominant *(WDD) when $H_{ii} \ge \sum_{j\ne i}|H_{ij}|$ on the* scalar *entries of the assembled H, not its $d\times d$ blocks; the left side is H_{ii}, not $|H_{ii}|$, so the inequality already forces a nonnegative diagonal. If H is symmetric and WDD, then $H \succeq 0$, and the Schur complement after each scalar pivot of the sweep is again WDD, so every pivot along the sweep is nonnegative.*

For the identity-coupled chain ($A_k = I$, $R_k = I$) with full-stage keep-outs ($\Pi_j = I$), WDD is exactly the per-stage condition $w_k \geq \lambda_k$, where w_k is the scalar goal weight ($W_k = w_k I$) and λ_k the total multiplier load on stage k: the goal weight dominates the keep-out multipliers. Under a projection Π_j the load falls only on the guarded coordinates, so WDD becomes the per-coordinate condition $w_k \geq \max_i \lambda_{k,i}$, with $\lambda_{k,i}$ the summed multipliers of the keep-outs guarding coordinate i of stage k. The stage condition $w_k \geq \lambda_k$ remains sufficient, and remains sharp whenever some coordinate is guarded by every keep-out on the stage — in particular for a single projected keep-out; it is strictly conservative exactly when the guarded sets of the keep-outs carrying positive *multipliers have no coordinate common to all of them, which, those sets being nonempty, forces at least two positive multipliers.*

Proof. A symmetric WDD matrix is PSD by Gershgorin (machine-checked; [30]). The Schur complement of a strictly diagonally dominant matrix is strictly diagonally dominant [7]; the weak case follows by perturbing the diagonal by $\varepsilon \to 0^+$, zero pivots having zero rows in the symmetric WDD case and being skipped, as in Lemma 1. Applying this one pivot at a time keeps every Schur complement WDD, hence every pivot nonnegative; $H \succeq 0$ itself is the Gershgorin step above. In the identity-coupled scalar specialization the whole check reduces to the local test $w_k \geq \lambda_k$ on each row of H — interior rows $[-1,\ 2 + w_k - \lambda_k,\ -1]$ and the two boundary rows $[\,1 + w_k - \lambda_k,\ -1\,]$ — with no block pivot S_k computed, which also avoids presuming the nonsingularity that S_k's recursion would need. □

WDD is sufficient, not necessary. It is also necessary on a boundary family: the full-stage identity-coupled free-endpoint chain has $H = T \otimes I_d$ with T the $N \times N$ path Laplacian plus $\mathrm{diag}(w_k - \lambda_k)$, and the path Laplacian annihilates the constant vector, so $\mathbf{1}^\top T \mathbf{1} = \sum_k (w_k - \lambda_k)$ (equivalently $\mathbf{1}^\top H \mathbf{1} = d \sum_k (w_k - \lambda_k)$) and hence, along any ray $c\,v$ of a fixed nonzero entrywise-nonnegative excess pattern v (uniform excess $w_k - \lambda_k \equiv c$ being one such), $H \succeq 0$ iff $c \geq 0$. Identity coupling is load-bearing — with $A_k \neq I$ a PSD H need not be WDD — whereas uniformity is not: what the argument needs is that the excesses do not differ in sign. With mixed signs necessity genuinely fails.

One keep-out per stage does not guarantee A1 either; the next example certifies a duality gap on exactly that geometry.

Example 1 (Certified duality gap)**.** A d=1, N=2 instance of (1) with one keep-out per stage, stage-coupling coefficient $-3/2$, and nonzero goals — in full: minimise $\left(x_1 + \frac{3}{2}x_0\right)^2 + (x_0 + 2)^2 + (x_1 + 2)^2$ subject to $\left(x_0 - \frac{1}{2}\right)^2 \geq 4$ and $x_1^2 \geq 4$ — has $J^\star = \frac{212}{13}$ at $x^\star = (-20/13,\ 2)$, by exact enumeration of the interior, edge and corner candidates in the four feasible branches. This enumeration is exhaustive because $P_0 \succ 0$ (its determinant is $17/4$) makes the objective strictly convex on each convex branch. Its exact level-one optimum and relaxation gap are

$$d_1^\star = \frac{24319}{1914}, \qquad J^\star - d_1^\star = \frac{89621}{24882}.$$

For the dual lower bound, take $\lambda = (\frac{14}{11}, \frac{25}{29})$ and $t = \frac{24319}{1914}$. Direct assembly gives

$$M(\lambda, t) = \frac{87}{44}\, z z^\top \succeq 0, \qquad z = (1, \frac{22}{29}, \frac{4}{3})^\top.$$

Here z is ordered as $(x_0, x_1, 1)$, the trailing-border layout of M. For the matching upper bound on the relaxation, use moments ordered as $(1, x_0, x_1)$:

$$Y = \begin{pmatrix} 1 & \frac{-97663}{111012} & \frac{-50353}{84216} \\ \frac{-97663}{111012} & \frac{79658}{27753} & \frac{-141311}{63162} \\ \frac{-50353}{84216} & \frac{-141311}{63162} & 4 \end{pmatrix}.$$

Writing $\mu = (Y_{01}, Y_{02})^\top$, the covariance of these moments is

$$\mathrm{Cov} = \frac{25833914015}{12323664144}\, vv^\top \succeq 0, \qquad v = (1, -29/22)^\top.$$

Hence $Y \succeq 0$. Exact substitution gives $E[f_1] = E[f_2] = 0$ and $E[f_0] = \frac{24319}{1914}$. Taking expectations of any globally nonnegative certificate polynomial $f_0 - \sum_j \lambda_j f_j - t$ therefore gives $t \le \frac{24319}{1914}$. The attaining dual certificate proves equality. `check_revision.py`, via `check_findings.exact_gap` (the `exact_gap` entry in `exact-checks.json`), verifies the Gram identity, the moment covariance, both constraint expectations and the primal branch enumeration, using exact rational arithmetic.

The same `check_revision.py` receipt also checks the primal KKT multiplier and curvature determinant, discharging the step from a gap to a failure of A1 rather than leaving it to the reader: $x^\star$ is rational (A2) and exactly one keep-out is active there, so LICQ is automatic and Theorem 1 would close the bracket if A1 held. It does not. A1 asks for *some* KKT pair with $H \succeq 0$, and any such witness is, by the convexity step in the proof of Theorem 1, a global minimiser: it would furnish a level-one certificate at $J^\star = \frac{212}{13}$, above the relaxation optimum $\frac{24319}{1914}$. At that unique optimum LICQ pins $\lambda^\star = 24/13$ and $H(\lambda^\star)$ has determinant $-7/4$, so $H(\lambda^\star) \not\succeq 0$. This is why aggregate-PSD (A1) is a real assumption and not automatic: under WDD no coupling is a gap source, and outside it any coupling — identity included — can break A1; what identity coupling buys is only that it consumes no WDD margin, which a general A_k need not preserve.

Hard dynamics beyond control elimination. When controls cannot be eliminated into (1) as in §6, retaining $x_{k+1} = A_k x_k + B_k u_k$ adds equalities and requires a different assembly. Lemma 3's criterion is stated for the displayed penalty-coupled assembly. The *closure* argument can still apply to an equality-constrained formulation: an affine equality contributes nothing to the Lagrangian curvature matrix and enters weak duality with a free multiplier, so wherever the analogue of aggregate-PSD holds at a rational KKT point the same constant-multiplier bracket closes. An exact receipt certifies both halves over $\mathbb{Q}$ on d=1 instances with one keep-out and one equality [30]. On the first, the best constant-multiplier bound is exactly 1 while $J^\star = 3/2$, a certified gap of $1/2$. The diagnosis, though, is Example 1's, not a statement about hard coupling: stationarity there recovers $\lambda^\star = 3/2$ against a unit curvature — both keep-out branches recover the same $\lambda^\star$, so no KKT pair escapes the verdict — and $H(\lambda^\star) = \mathrm{diag}(-1/2, 1, 1)$ is not PSD, so A1 fails, with A2 and LICQ intact. On its sibling — the same variables and the same equality, with the goal moved and the keep-out radius doubled — stationarity recovers $\lambda^\star = 3/4$ and $H(\lambda^\star) = \mathrm{diag}(1/4, 1, 1) \succeq 0$, so A1 holds and the constant-multiplier bracket closes exactly at $t^\star = J^\star = 9/4$: the same trusted dense M-test accepts $t^\star$ and refuses $t^\star + 10^{-6}$. The exhibited gap therefore measures A1, as Example 1's does, and no conclusion about Theorem 1 follows from it.

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